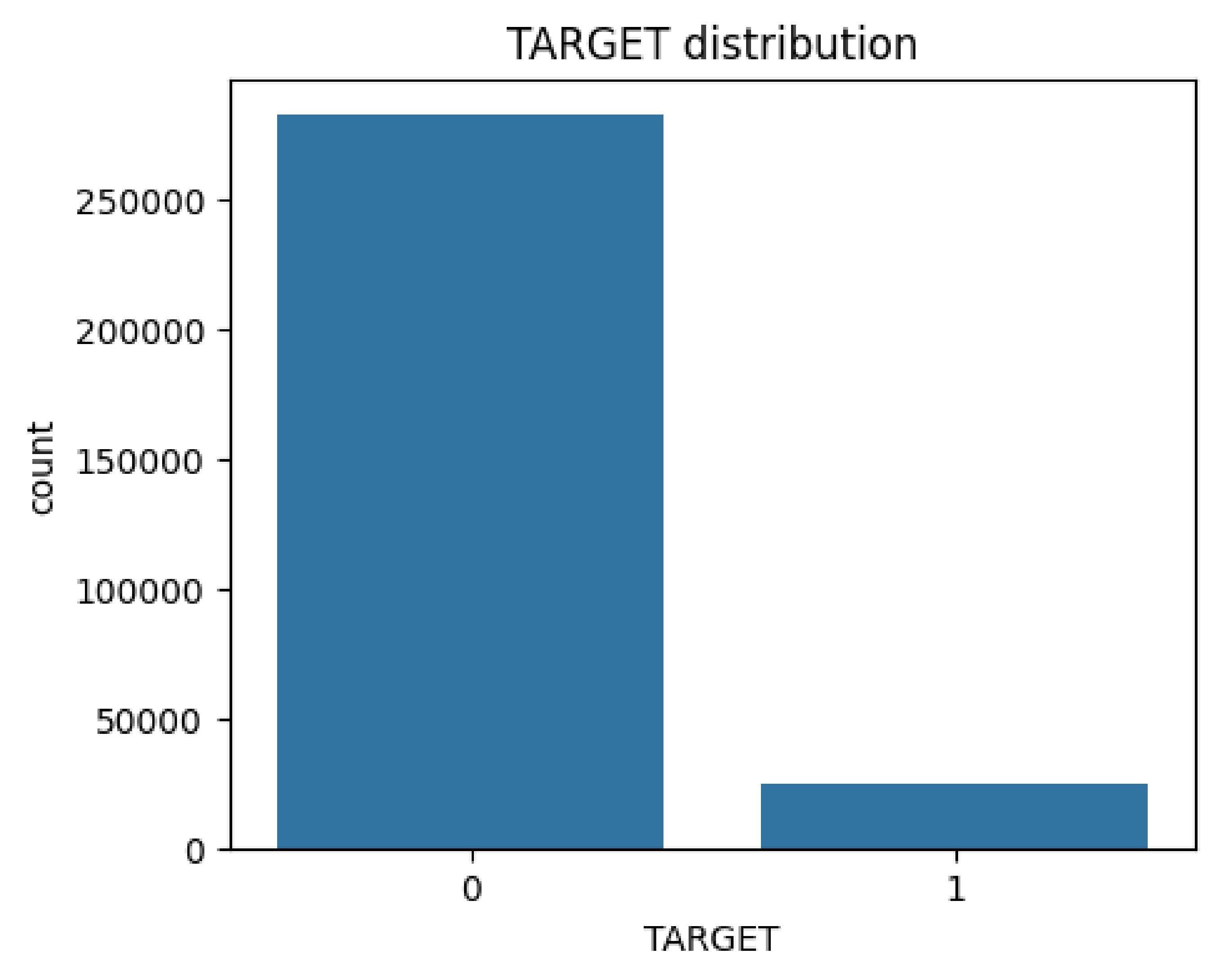
EDA CASE STUDY

OVERVIEW

An overview of the information attained from maniuplating the data provided.We look at the different demographics provided to see how those characteristics affect the target (0= no toruble paying back loan, 1= trouble paying back loan) of the loan.

NUMBER OF DEFAULTS

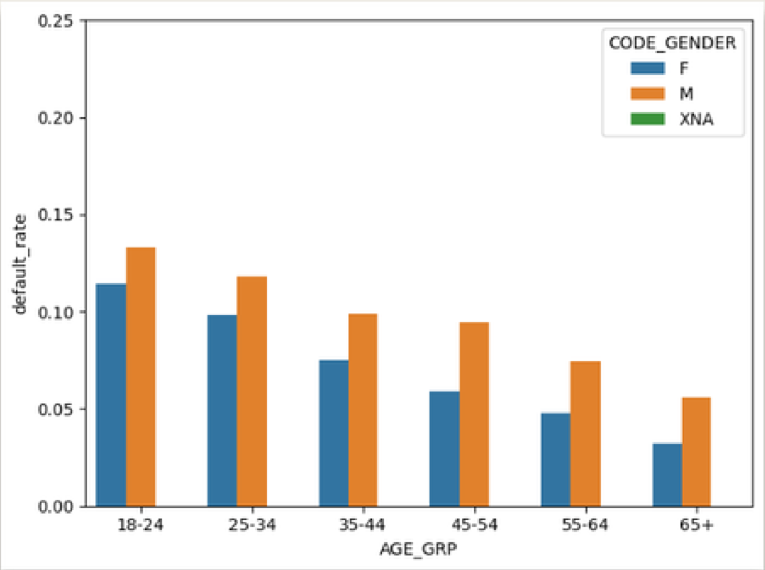
The amount of people in the data given who have had no troubles paying their loans is 282686, the number of people who have had difficulties paying is 24825. Meaning only 0.0807 or 8% of people have trouble paying back the loans provided to them. Showing the company has good practices in deciding who gets approved for loans as most of them repay the loans with no difficulties.



CHARACTERISTICS OF BORROWERS

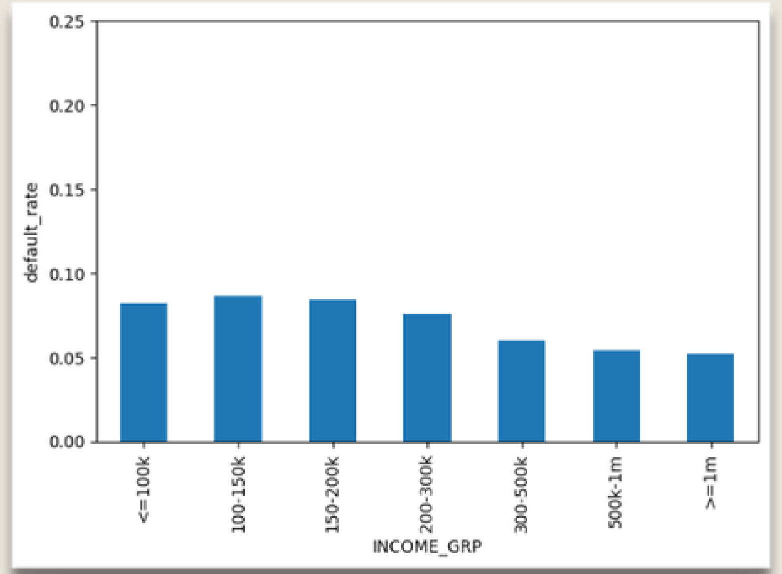
Gender

The distributions shows male applicants have a higher default rate than female applicants regardless of age . This suggests that gender may play a role in the ability to repay the loan. The company should take this into consideration while still be fair when deciding on which applicants to approve.



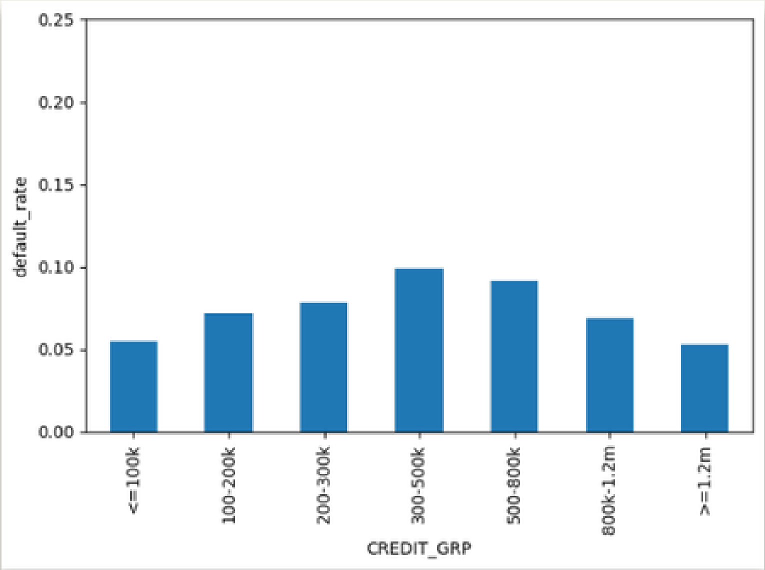
Income

The graph shows the income distribution across the applicants. We can conclude that applicants in the 100k-150k income range are more likely to have trouble paying back their loans. Though they have the highest default rate there isnt a huge difference compared to other income brackets showing the company does a good job looking at debt to income ratios.



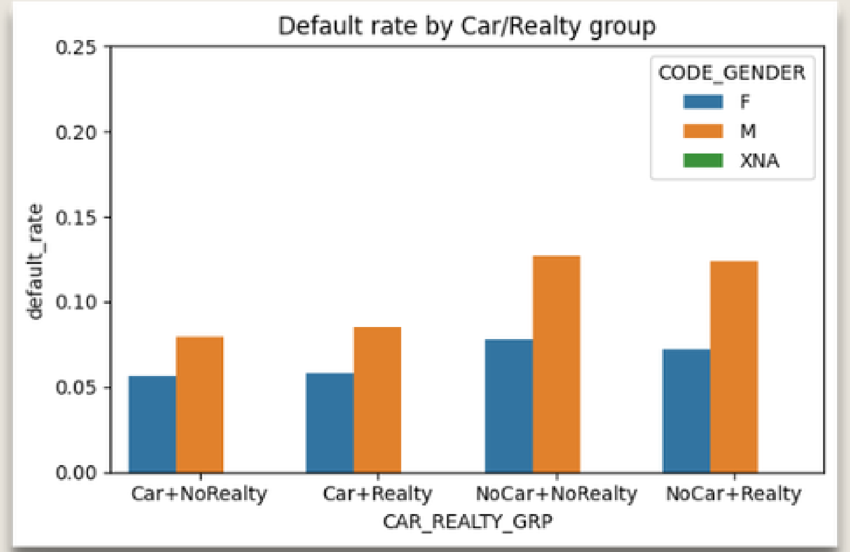
Credit

This shows the default rate based upon the ranges of credit that was given in the loan. The company can use this to analyse the credit amount that exposes them to the most risk, so that they can make more informed decisions on whether or not to approve loans of certain amounts.

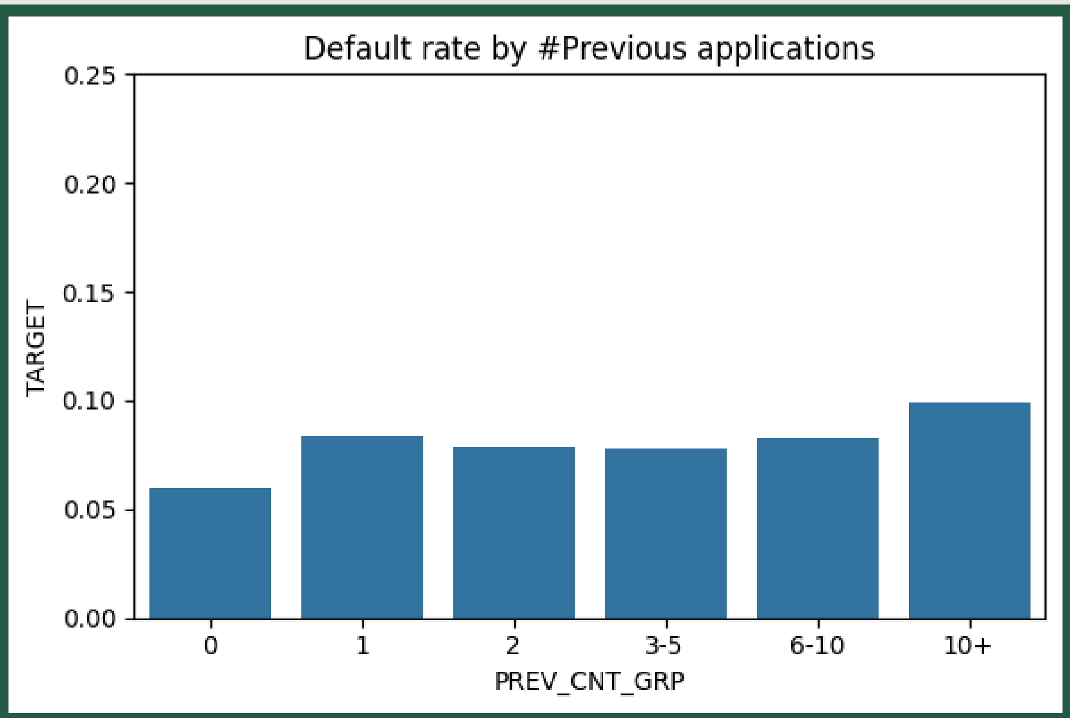


Cars and reality

This shows the default rate based on if the applicant owns a car and/or real estate. The data indicates that applicants with no car or real estate have a higher default rate compared to those with assets. This suggests that ownership of assets is something that has to be considered in the riskiness of loaning to applicants.



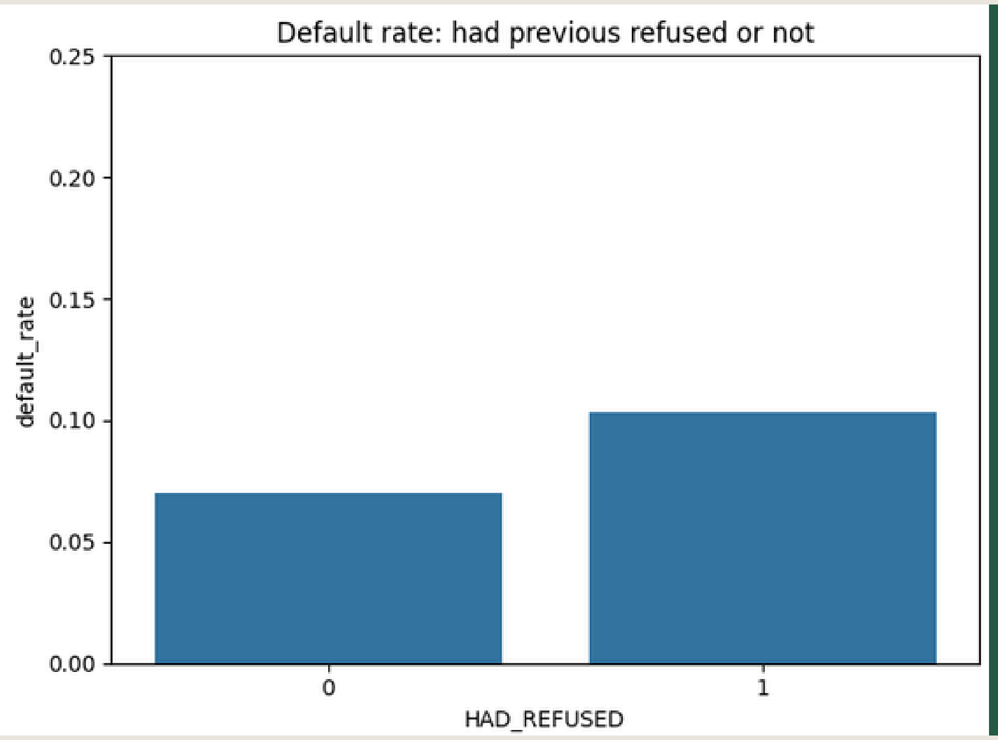
PREVIOUS APPLICATIONS

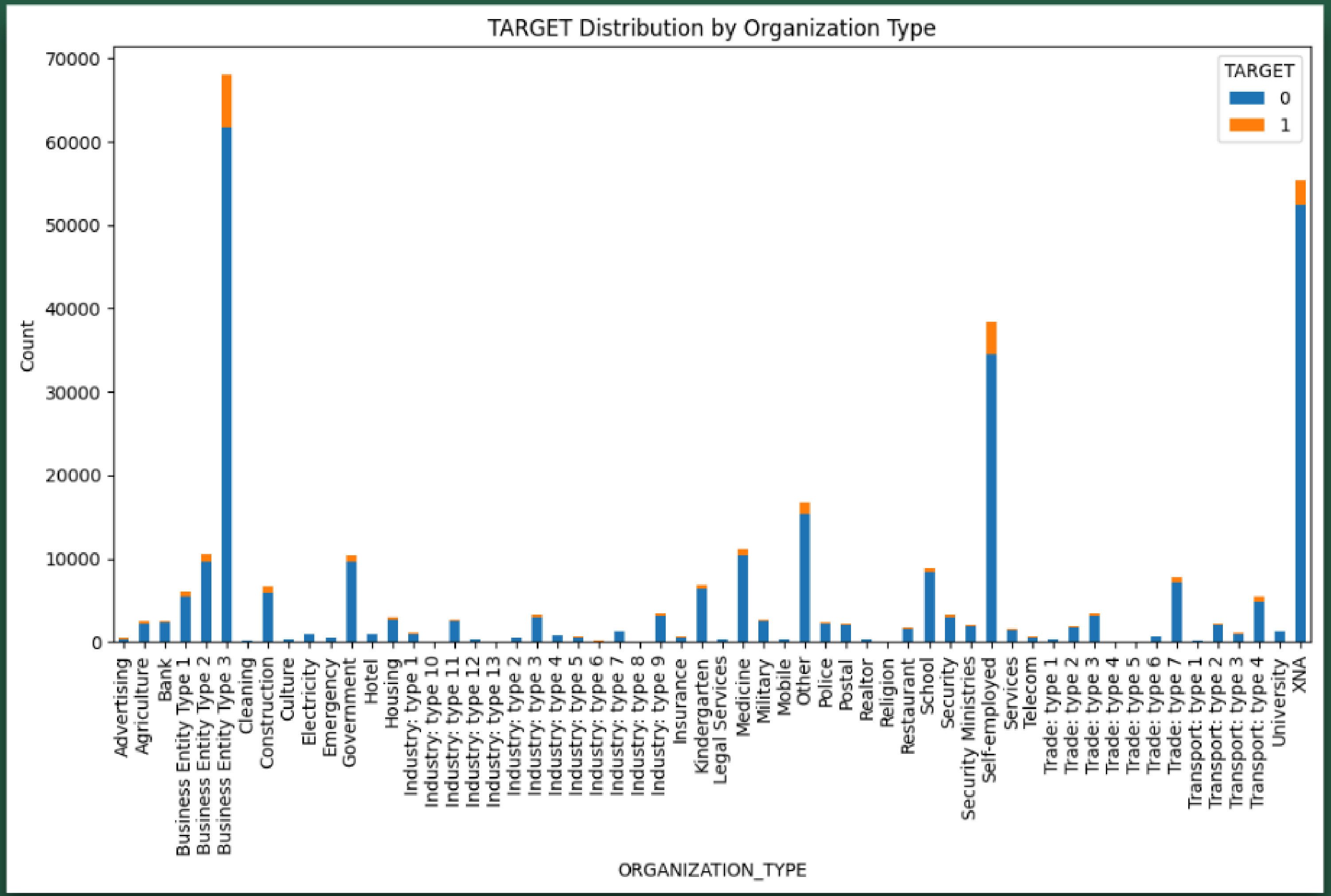


- Visualizes the default rate based on how many previous applications have been put in.
- Shows that the company probably shouldn't loan to people with 10+ previous applications as they have the highest default rates.

PREVIOUSLY REJECTED

- Shows the target rate based on if an applicant has been previously rejected from getting a loan.
- The company most likely shouldn’t give loans to people who have perviously been rejected as they have a higher default rate.





DEFAULT BASED ON JOBS TYPE

Graph Overview

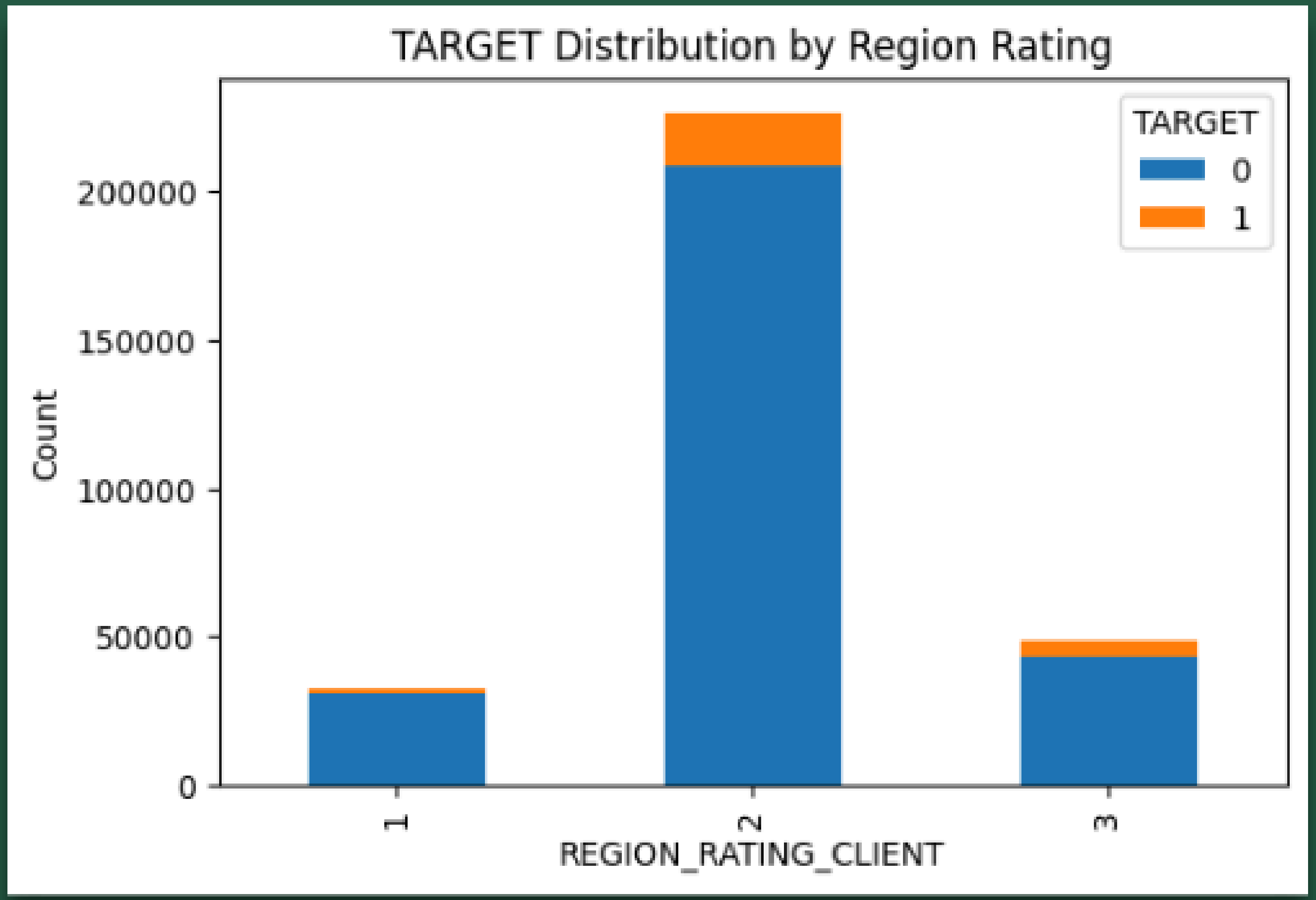
The graph shows the amount of people who have applied from each industry and whether or not they have difficulties making their payments

Information from graph

Most people who apply have no difficulties paying their loans back regardless of the industry they are in. Showing that the type of organization a person works at may not actually matter on whether a person will default on their loan.

REGION

- Looks at default rate aka target rate and how that is affected by the rating the company gives the region they live in
- Shows that the region the applicant lives in and the rating the company provides based on that region may not be applicable in determining if a client will default or not as majority of the applicants regardless of the region pay back their loans
- However the data shows that the company is more preferential to giving out loans to people who live in the regions that they rate a 2.



DATA ON THE LOAN

NAME_CONTRACT_TYPE	AMT_CREDIT	AMT_GOODS_PRICE	TARGET
Cash loans	45000	45000	0.052174
	47970	45000	0.041284
	48519	40500	0
	49455	45000	0.052632
	49500	49500	0
...	...	...	...
Revolving loans	1575000	1575000	0
	1687500	1687500	0
	1800000	1800000	0
	2025000	2025000	0
	2250000	2250000	0

Overview

- Shows the amount of credit the applicant received along with the price of the good they wan to purchase. The Target shows the default probability of an applicant with those exact contract details.
- However since credit and price are continuous variables there may only be one or two people in a group so it doesn't tell us much.

Action

- The company should analyze the default percentage across credit amounts

WAS THE CLIENT REACHABLE ?

Flag_Count_ Mobile	TARGET	COUNT	% as a decimal
0	0	529	.922
	1	45	.078
1	0	282157	.919
	1	24780	.081

This shows whether or no the company was able to reach the client on their mobile phone with 1= yes and 0= no. The table then breaks down the amount of poeple who have defaulted and answered the phone and the amount who have ant and answered the phone and vice versa. From this we can see that despite the differnece in amounts whether or not the client answered the phone the default rate is about 8% for each group meaning that this information doesnt tell us much in figuring out if a client will default on their loan.

THANK YOU